

Range review: what to discontinue, and which listings to fix

To Head of Merchandising

From Andres Perez, analytics

Data Every invoice line, December 2009 to November 2011 (UCI Online Retail II)

RECOMMENDATION

1. **Discontinue 906 products (22% of the range).** These are low-revenue products that fewer than three of our key accounts buy. Chosen on 2010 data and tested on the following year, they would have put at most **0.6% of next-year revenue** at risk (\$87k of \$14.72M; 95% interval 0.5–0.7%).
2. **Do not cut the standard "class C" tail.** It is 1,969 products, but it would have put 3.1% of revenue at risk (2.7–3.5%). That is over the 3% limit we set before looking at the results, and it touches 60% of customers.
3. **Send the 50 products with the highest cancellation rates for a listing review, and refresh the list every year.** The products with the most cancellations in one year cancelled at 4.6× the rate of other products in the next (95% interval 3.6–5.8×).

HOW EACH CUT LIST HELD UP ON THE FOLLOWING YEAR

Cut list (chosen on 2010 data)	Products	Next-year revenue at risk	Picks removed	Customers affected
Class C tail (bottom 5% of revenue), new launches protected	1,969	3.1%	7.9%	60%
Class C tail, no protection for new launches	2,087	4.7%	10.2%	71%
Fewest orders (same size as the first list)	1,969	3.5%	6.3%	61%
Class C tail, minus products bought by 3+ key accounts	906	0.6%	1.4%	25%
A random set the size of the first list	1,969	36%	–	–

Amounts in US dollars, converted from British pounds at \$1.5774 per £1. Key accounts are the top 10% of customers by 2010 revenue. Picks are the share of order lines, a proxy for warehouse workload. "Revenue at risk" assumes no customer switches to a similar product.

WHY THE CLASS C CUT FAILS, AND WHAT FIXES IT

- **The tail is long:** half the 2010 range (2,087 products) earned 5% of revenue.
- **But it isn't dead weight:** 81% of the class C tail's at-risk revenue sat in products that at least three key accounts were buying. Several grew sharply the next year; a rosette went from \$652 to \$5,687. Protecting those products keeps nearly all the revenue and still removes 906 listings.
- **New products need protection:** without it, the cut list held twice as many products that became next year's strong sellers (123 against 62).
- **Much of the tail drops out anyway:** 368 of the 906 recommended products had already stopped selling the following year. The range turns over about a fifth of its products each year without a formal rule; this makes the turnover deliberate.

LIMITS

- **Revenue only.** The data has no product cost, margin or stock levels. The savings from a smaller range (stock, storage, catalogue upkeep) are real but not measured here. The next step is to add margin and holding cost before the final list is signed off.
- **One year tested:** the test covers a single year-to-year transition.
- **Selection:** the recommended list was one of four rules written down before the test. Picking it after seeing all four results adds some selection risk, but its result sits far inside the limit.

Method, code and full results: github.com/Andresperez397/retail-operations-case-study. The analysis plan was committed before any next-year result was computed.